



Exploratory Data Analysis(EDA) on Retail Sales Data

Tools used:

- Google Colab
- Excel
- Power BI
- GitHub

PHASE 1: Understanding the Dataset:

- There are 1000 rows and 9 columns
- Columns names include:

<u>Numerical columns:</u>	<u>Categorical columns:</u>
1. Transactional ID 2. Date 3. Age 4. Quantity 5. Price per unit 6. Total amount	7. Customer ID 8. Gender 9. Product Category

PHASE 2: Data Loading and Cleaning

I've initially loaded it into Google Colab and have done the cleaning by

- removing duplicates
- handling missing values
- formatting the data types

PHASE 3: Descriptive Statistics

1. Identified numerical columns

- Transactional ID
- Date
- Age
- Quantity
- Price per unit
- Total amount

2. Generated summary statistics like

Mean(average sales)	456.0
Median(middle value)	135
Mode(most frequent value)	50
Standard deviation(data spread)	559.997631555123

PHASE 4: TIME SERIES ANALYSIS

1. Ensured **date column** is in proper **format** which is mandatory.
2. Extracted **time features**
 - decomposed into **year, month, and day** components for time-based analysis.

Monthly Sales Analysis:

1. **May** month had the highest sales
2. **September** had the lowest
3. There's no repeating pattern.

Yearly Sales Analysis:

1. gave a conclusion that the sales were in the **dropping pattern** with a steep line.

Daily Sales Analysis:

1. **Saturday** had the highest sales
2. **Thursday** had the lowest

Weekly Sales Analysis:

1. **Weekends** had more sales compared to the week days.

PHASE 5: CUSTOMER & PRODUCT ANALYSIS

1. **Electronics** generated highest revenue with dominant sales, compared to the other two.
2. **Females** were the top contributors on the revenue than males.
3. Identified **high-value customers** on basis of their average amount spent.
4. Customers were **segmented** based on spending behavior into high-value and low-value groups.

PHASE 6: DATA VISUALIZATION

- ***In colab***

Imported Visualization Libraries: Matplotlib and Seaborn

1. ***CATEGORY-WISE SALES***

The bar chart shows that **Electronics** contributed the highest sales, while **Beauty** has the lowest performance.

2. ***MONTHLY SALES TREND***

The line graph highlights trends and seasonal patterns in sales over time and concluded that May was the highest sales month.

3. ***SALES DISTRIBUTION***

The histogram shows the distribution of sales values, indicating skewness and variability. It shows the max value at the range of 0-250.

4. ***CATEGORY SHARE***

The pie chart shows percentage contribution of each category.

Electronics- 34.4%

Clothing-34.1%

Beauty-31.5%

5. *BOXPLOT*

The boxplot looks for outliers and spread of sales data and concluded that there were no outliers.

- ***In Power BI*** created graphs on
 1. Year by Total Amount
 2. Month by Total Amount
 3. Day by Total Amount
 4. Product Category by Total Amount
 5. Age by Total Amount

PHASE 7: Recommendations(Insights):

- The business relies heavily on a ***few high-value transactions***, indicating potential dependence on premium customers.
- Business performance is influenced by time-based demand pattern.
- Strong relationship between sales and profit. A positive correlation between sales and profit indicates that increasing sales generally leads to increased profitability.
- Category analysis indicates that not all product categories contribute equally to total sales. Electronics were the highest sellers.
- Revenue is concentrated among a few customers, Additionally, many customers make fewer or smaller purchases
- Analysis of purchase frequency shows that some customers make repeated purchases, while others are one-time buyers.